3.3 - Increasing Functions

Goals: Connect properties of the derivative to the idea of increasing functions and extrema
Before Class Videos: No video for this section; consult Section 3.3 of APEX Calculus to review.

Definition:
$f(x)$ increasing on interval I if for all $a < b$ in I $f(a) < f(b)$ in I.

Definition:
$f(x)$ decreasing on interval I if for all $a < b$ in I $f(a) > f(b)$ in I.

Theorem:
If $f'(x) > 0$ on $[a, b]$, then $f(x)$ increasing on $[a, b]$.

Example 1: Decide the intervals on which $f(x) = 2x^3 - 3x^2 - 12x$ is increasing and decreasing. Sketch a very loose graph of the function.

Theorem (1st Derivative Test):

Let c be a critical number of f, and f both continuous at c and differentiable near c.

1. $f'(x)$ changes sign from $+$ to $-$ then $f(c)$ is a relative max.
2. $f'(x)$ changes sign from $-$ to $+$ then $f(c)$ is a relative min.
3. If there is no change in sign, then $f(c)$ is not a relative extremum

Example 2: Classify all critical numbers of

$$f'(x) = \frac{e^{-2x}}{20}(x-1)(x+1)^2.$$

Example 3: Classify all critical numbers of

$$g'(x) = \frac{(x+4)(x-1)^2}{x-2},$$

Example 4: Sketch graph of function such that:

- V.A. at $x = 1$;
- $f'(x) < 0$ on $(1, \infty)$;
- $f'(x) > 0$ on $(-1, 1)$;
- minimum at $x = -1$;
- $\lim_{x \to -\infty} f(x) = 0$;
- continuous elsewhere.

Example 5: Determine intervals where $f(x) = \cos x + \sin x$ is increasing and decreasingon $[0, 2\pi]$.

Suggested after class problems from APEX: Section 3.3: 15,21